ASPEN INTEGRA

ENGINEERING WHITEPAPER

Glass-former selection in the EWG2.5 formulation model

Formal exposition of the constrained formulation procedure, with a complete worked example on the Batch 3 waste

Aspen Integra

November 3, 2026 | AI-WP-1013 rev 2.1

This paper presents the formulation procedure by which the software in ewg_app/ewg converts a waste oxide composition into a specified mass of each glass-forming chemical (GFC). Selection is performed by constrained numerical optimization rather than by reference to a recipe library: each of ten candidate GFCs is a continuous decision variable, the waste itself is one further variable, and the recipe is the constrained maximum of waste loading under the property models of PNNL-35884 Rev. 1. The complete calculation is exhibited on a single waste composition, the Batch 3 aluminous-sodic waste of Table 3-3, and each numerical result is reproducible from the code with the commands given in the text. Section Results follows the example waste through every stage of the calculation; Section Methodology states the governing equations and algorithm formally.

Executive summary

The program determines a glass-former recipe by solving for the composition of highest admissible waste loading, subject to every glass property limit of the underlying basis. No preset recipes are consulted. The ten candidate glass-forming chemicals (GFCs) enter the calculation as continuous decision variables; the delivered recipe comprises exactly those GFCs whose decision variables remain nonzero at the constrained optimum. A GFC that offers no benefit for a particular waste returns at zero and is not procured.

Three numerical results characterize the method as applied to the Batch 3 waste. The solver attains a waste loading of 54.73 wt% with six formers in the recipe: silica, boric acid, lithium carbonate, zincite, zircon, and vanadia. The solution is bound at convergence by the PCT durability limit adjusted inward by its predicted uncertainty (margin 5 x 10^-10), with the 1150-degree viscosity ceiling and the NSiAl model-validity ceiling also resting on their respective limits. Total raw chemical demand is 0.5824 kg per kg of glass.

PRINCIPAL FINDINGS

(1) Recipe selection is a byproduct of maximizing waste loading subject to the Table 3-1 property constraints; no recipe library is consulted. (2) The waste coordinate of the decision vector is itself the waste loading, so the objective reduces to maximizing a single variable, x[0]. (3) V2O5 is admissible only when the waste carries more than 0.5 wt% SO3; chromia, when admissible, is capped at 0.6 wt% of the glass. (4) All margins are enforced inward by predicted uncertainty, so the delivered recipes satisfy the published limits with allowance to spare. (5) Raw chemical masses are recovered after the solve by division of each oxide share by the chemical's oxide yield.

Introduction

Copyright 2026 Aspen Integra

A joule-heated melter converts a slurry of calcined waste and glass-forming chemicals into borosilicate glass. The chemicals, principally silica sand, boric acid, lithium carbonate, soda ash, and several minerals, exist to reshape the oxide inventory delivered by the waste into a glass that can be poured at operating temperature and that passes the durability acceptance tests. Waste compositions across the Hanford tank farm vary widely, from wastes that are half alumina to wastes that carry 61% soda, and a fixed recipe would either consume excess former or fail acceptance limits on most of them. The recipe must therefore be computed per waste.

The software examined here comprises two modules: ewg/gfc.py, which defines the GFC library, and ewg/optimize.py, which implements the solver. Both depend on ewg/properties.py, which encodes the property models of PNNL-35884 Rev. 1. The paper addresses one question: given a waste oxide vector, by what computational procedure does the code determine which formers to include, and in what quantities? Section Results executes the full calculation on one waste and may be read independently. Section Methodology states the governing equations and algorithm formally.

Results: the Batch 3 waste followed through the calculation

Batch 3 is one of three example wastes distributed with the code (Table 3-3 of PNNL-35884) and serves as the anchor case of this analysis: the documented Batch 3 recipe reproduces Table 3-5 of the referenced paper to four decimal places under the conventions stated below, which constitutes the verification of the entire pathway.

Step 1. The waste composition supplied to the optimizer

The input is a dictionary of oxide mass fractions summing to unity, one entry per oxide. Batch 3 is dominated by three oxides that jointly account for 86.4% of the waste: alumina at 50.61%, soda at 29.06%, and ferric oxide at 6.60%. The remainder is present in trace quantities. This composition governs the subsequent behavior of the optimizer: the waste already carries substantial Na2O, so the solver is under little pressure to procure soda ash, and the large alumina inventory must be diluted into a glass that satisfies the viscosity and compositional windows.

Copyright 2026 Aspen Integra

Oxide	Mass fraction	Interpretation
Al2O3	0.50614	Half of the waste is alumina; the NSiAl validity ceiling will constrain dilution.
Na2O	0.29061	Soda enters at no cost from the waste; soda ash remains unused in the final recipe.
Fe2O3	0.06604	Iron is carried through the glass and is not a GFC ingredient in this case.
SiO2	0.05944	The network-former inventory is small; silica sand supplies most of it.
UO3	0.00884	Carried as oxide and tracked in the mass balance.
ThO2	0.00803	As above.
CaO	0.00786	Trace; wollastonite is not required.
P2O5	0.00711	Compared against the phosphate limit; remains far below it.
SO3	0.00517	Above the 0.5 wt% gate, so V2O5 is presented to the solver.
All others (25 oxides)	0.04076	Trace actinides, PGMs, halogens, REEs, NiO, ZrO2; summed from gfc.py.
Total	1.00000	Normalized to total waste oxide in the waste row of the matrix.

Table 1. Batch 3 waste composition as supplied to the optimizer.

Step 2. The candidate formers and their oxide yields

Each glass-forming chemical is stored as the mass of useful oxide delivered per unit mass of as-fired chemical. Pure oxides yield 1.0. Minerals divide their mass between two oxides in proportion to molecular weight. Hydrated and carbonate chemicals surrender a fraction of their mass to volatiles: boric acid delivers 0.5630 kg of B2O3 per kg because 3 H2O leave the melt per 2 H3BO3, and soda ash delivers 0.5850 kg of Na2O per kg because the carbonate evolves as CO2. Table 2 lists the library available to the solver.

Copyright 2026 Aspen Integra

GFC	Raw chemistry	Oxides delivered (kg/kg as-fired)
Silica	SiO2	SiO2: 1.0
Boric acid	2 H3BO3 -> B2O3 + 3 H2O	B2O3: 0.5630
Lithium carbonate	Li2CO3 -> Li2O + CO2	Li2O: 0.4043
Soda ash	Na2CO3 -> Na2O + CO2	Na2O: 0.5850
Kyanite	Al2SiO5 -> Al2O3 + SiO2	Al2O3: 0.6283; SiO2: 0.3717
Wollastonite	CaSiO3 -> CaO + SiO2	CaO: 0.4828; SiO2: 0.5172
Zincite	ZnO	ZnO: 1.0
Zircon	ZrSiO4 -> ZrO2 + SiO2	ZrO2: 0.6722; SiO2: 0.3278
Chromia	Cr2O3	Cr2O3: 1.0 (capped at 0.6 wt% of glass)
Vanadia	V2O5	V2O5: 1.0 (admissible only when waste SO3 > 0.5 wt%)

Table 2. The ten candidate glass-forming chemicals and their as-fired oxide yields (gfc.py).

Step 3. Decision variables and objective function

The solver operates in eleven variables: x[0] for the waste and x[1] through x[10] for the GFCs. Each x[k] is the mass fraction of that component's oxide in the final glass, and the eleven coordinates are subject to a summation-to-unity equality constraint. The objective is a single expression, the maximization of x[0], the waste-oxide fraction of the glass. No other quantity is optimized. Recipe selection follows from feasibility alone: a former acquires a nonzero coordinate only where its inclusion permits the model to admit more waste while every property limit is satisfied.

Step 4. Property constraints that shape the recipe

At each trial point the code combines the eleven oxide rows by a single matrix product (g = x M), renormalizes to total oxide mass, and evaluates the property models. Fourteen inequality families are enforced, each on the per-model basis of the referenced paper: PCT durability, melt viscosity at 1100 and 1150 degrees, electrical conductivity at both temperatures, the k1208 crystal-fraction proxy, the phosphate limit, the Na-Li immiscibility margin, the alkali index NALK window, the NSiAl window, sulfate solubility, and the per-component composition bounds of Table 3-2. Two aspects of the constraint set require emphasis because they perform the binding in the Batch 3 case.

First, each margin is adjusted inward by its predicted uncertainty Upred. The durability limit of 6.4368 g/m2 is imposed as 6.4368 minus the 4.644 g/m2 predicted-basis allowance; the solver thereby targets a glass that the models certify as durable, and not merely a glass the models cannot exclude. Second, the binary rules reside in the variable bounds rather than in the constraint set: for Batch 3 the waste SO3 content of 0.517 wt% lies above the 0.5% gate, so the

V2O5 bound remains open at (0, 1) and the solver may admit vanadia. Had the waste been below the gate, the bound would be fixed at (0, 0) and vanadia would be invisible to the optimization irrespective of its potential benefit. The 0.6 wt% chromia ceiling is likewise imposed as a bound on the GFC variable itself.

Step 5. The solution and the delivered recipe

The solver is SLSQP from scipy.optimize, executed from five starting points (waste fractions 0.40, 0.20, 0.10, 0.05, and 0.30) with retention of the best feasible result. Multiple starts are required because a single start may place the line search in an infeasible region from which it cannot recover; Batch 17, with 61.5% soda in the waste, fails in precisely this manner. For Batch 3 the first start succeeds, and the solution reproduces from a fresh interpreter.

Component	Oxide share of glass, x[k]	Raw chemical, kg per kg glass
Waste	0.5473	not applicable (defines the waste loading)
Silica	0.2143	0.2143
Boric acid	0.0856	0.1521 (oxide share / 0.5630 yield)
Lithium carbonate	0.0430	0.1062
Vanadia (V2O5)	0.0507	0.0507
Zincite (ZnO)	0.0399	0.0399
Zircon	0.0192	0.0192
Kyanite	0	not selected
Wollastonite	0	not selected
Soda ash	0	not selected
Chromia	0	not selected
Total GFC demand	0.4527 (sum of above)	0.5824 kg per kg glass

Table 3. The Batch 3 recipe: the formers selected by the solver and their quantities. Raw mass equals the oxide share divided by the yield of Table 2; the boric acid entry verifies as 0.0856 / 0.5630 = 0.1521.

Six formers receive nonzero coordinates, and the selection logic admits a direct reading in three terms. Silica is the network former; with only 5.9% SiO2 in the waste, it receives a large share of the glass (21.4%) in order to satisfy the silica-alumina window. Boro and lithia condition the melt, B2O3 contributing durability and viscosity control, Li2O viscosity control, each within the tolerance of the property models. Vanadia appears only because the waste lies slightly above the sulfate gate; it increases the sulfate capacity of the melt. Soda ash is not required: the waste's own soda fully supplies the 15.9% Na2O carried by the final glass. Kyanite is likewise not selected, since the waste already arrives half-alumina and the NSiAl ceiling acts in the opposite direction.

Step 6. The constraints in force at the optimum

The waste loading of 0.5473 is not a value supplied by convention; it is the boundary of the feasible region. Table 4 evaluates every constraint family at the converged recipe. Three margins are essentially zero: the PCT durability limit adjusted for its uncertainty allowance (5 x 10^-10), the upper NSiAl validity ceiling (exact by construction), and the conductivity at 1200 degrees with its uncertainty allowance included. An increase of the waste coordinate above 0.5473 drives the first of these negative. The recipe is accordingly the vertex at which the durability bound meets the validity bounds.

Property (basis)	Value	Hard limit	Inward-adjusted limit	Margin
PCT durability, g/m2	1.7927	<= 6.4368	<= 1.7927 (allow 4.644)	0.0000 (BINDING)
eta at 1150 C, Pa s	5.700	4.0 to 6.0	4.300 to 5.700	0.0001 (binding)
NSiAl (SiO2 + 1.697 Al2O3)	0.7236	0.457 to 0.7236	identical	0.0000 (BINDING)
Conductivity at 1200 C, S/cm	0.6681	<= 0.7	<= 0.6681 (allow 0.032)	0.0000 (BINDING)
Conductivity at 1100 C, S/cm	0.5036	>= 0.1	>= 0.1269	0.3767
eta at 1100 C, Pa s	9.419	<= 15.0	<= 14.472	5.053
NALK (Na2O + 0.66 K2O + 2.07 Li2O)	0.2496	0.1381 to 0.2743	identical	0.0247 to hi
Na-Li immiscibility margin	0.4225	>= 0.2	identical	0.2225
k1208 crystal proxy	0.0137	<= 0.04	identical	0.0263
Phosphate limit	0.0124	<= 0.24	identical	0.2276
Sulfate SO3 in glass, wt%	0.283	<= 0.783 (3TS - 0.33)	<= 0.564 (allow 0.219)	0.281
Na2O in glass, wt%	15.91	4.1 to 27.0 (composition bounds)	identical	0.111 to hi
Mass closure	0.99943 of 1.0	not applicable (physical)	not applicable	volatile shortfall 0.057%

Table 4. Every constraint family at the converged Batch 3 recipe. Margins are quoted relative to the limits the solver actually enforces, with uncertainty folded in where the models provide it.

Copyright 2026 Aspen Integra

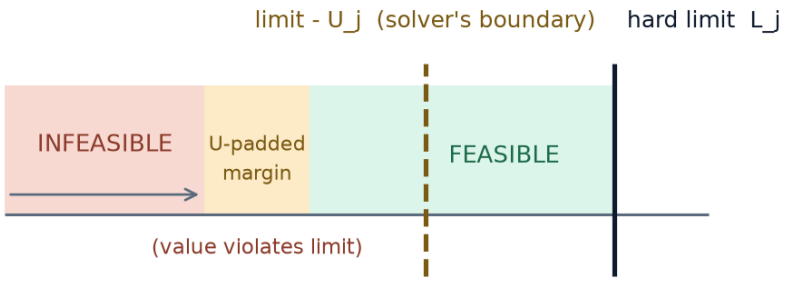


Figure 1. Structure of an enforced margin. The solver treats the inward-adjusted boundary (limit minus the predicted-uncertainty allowance) as the feasibility frontier; the hard limit stands one allowance further out.

The glass produced by the recipe

Multiplication of the eleven decision variables by the contribution matrix yields the final glass oxide vector, the quantity the entire calculation is arranged to control.

Oxide	Origin of the mass	wt% of glass
Al2O3	entirely waste	27.72
SiO2	silica sand 21.43; waste 3.25; zircon 0.64	25.33
Na2O	entirely waste	15.91
B2O3	boric acid	8.60
V2O5	vanadia	5.08
Li2O	lithium carbonate	4.30
ZnO	zincite	4.00
Fe2O3	entirely waste	3.62
ZrO2	zircon 1.29; waste 0.24	1.54
UO3, ThO2, CaO, P2O5, MnO, minor	entirely waste, each < 0.5%	1.92 combined

Table 5. The predicted final glass for Batch 3, as returned by evaluate_mix() at the solution point.

Reproduction of the numerical results

The following commands, executed from the ewg_app directory, print the waste loading, the recipe, and the constraint evaluation reported above.

Reproduction, Python, from the ewg_app directory

from ewg.gfc import EXAMPLE_WASTES / from ewg.optimize import formulate / r = formulate(EXAMPLE_WASTES["Batch3"]) / print(r["waste_loading"], r["gfc_raw_mass_fraction"]) / print(r["evaluation"].constraints)

The output begins with 0.5473 followed by the six raw masses of Table 3. For comparison, the same call on Batch 17, the high-soda, fluoride-rich waste, returns a waste loading of 22.8% with kyanite, wollastonite, and soda ash selected and vanadia again present; Batch 45 returns 40.3% with the capped chromia in the recipe. Each waste composition produces a distinct recipe.

Methodology

Figure 4 depicts the complete calculation as a data flow. The waste vector and the GFC yield vectors are assembled into the composition matrix; the optimizer adjusts the decision vector within its bounds and gates; each trial vector is mixed and evaluated against the property models; and a feasible optimum, once reached, is converted into raw chemical demand by yield division. The feedback edge denotes the solver's handling of infeasible trial points: the property evaluation drives the next adjustment of x.

Statement of the formulation problem

Let w denote the oxide mass-fraction vector of the waste, normalized to total waste oxide, and let e_k denote the oxide yield vector of glass-forming chemical k. The composition matrix M stacks the waste row over the ten GFC rows, all on a common 48-oxide column basis (FULL_OXIDES in optimize.py). The decision vector is x = (x_waste, x_1, ..., x_10), whose coordinates are oxide mass fractions of the final glass. The selection program is stated in Equation (1): maximize the waste coordinate subject to summation, nonnegativity, and the property constraints of PNNL-35884 Table 3-1 evaluated on the renormalized glass vector.

$$\max_{x \geq 0} x_{\text{waste}} \quad \text{subject to} \quad \sum_{k=0}^{10} x_k = 1, \quad g = \frac{xM}{\mathbf{1}^{\top}(xM)}, \quad s_j(g) + m_j(g) \geq 0,\ j = 1 \ldots 14$$

Figure 2. Equation (1). The glass-former selection program. s_j(g) denotes the constraint residual of family j and m_j(g) its variable bound; both are evaluated with the uncertainty allowances of Section Uncertainty handling.

The waste coordinate x_waste plays a dual role: it is both a component of the mixture and the waste loading of the glass, which is the quantity of operational interest. The solver therefore optimizes the operating objective directly rather than through a derived expression.

Copyright 2026 Aspen Integra

$$M = [\, w^{\top} ; e_1^{\top} ; \ldots ; e_{10}^{\top} \,] \quad \rightarrow 11 \text{ rows (waste + 10 GFCs)} \times 48 \text{ oxide columns}$$
$$w = w / \Sigma w \quad \text{(waste row, normalized)} \qquad e_k = \text{as-fired oxide split of GFC k (gfc.py)}$$

Figure 3. Equation (2). Construction of the composition matrix M from the normalized waste vector w and the ten GFC oxide-yield vectors e_k.

The mixing convention and its verification

The GFC rows of M are the relative oxide splits of Table 2, normalized to unit sum, and their as-fired losses (hydration water, carbonate-derived CO2) are deferred to the raw-demand recovery. The waste row is normalized to total waste oxide, and its approximately 1.2% shortfall relative to unity is retained as the physical volatile fraction. This convention is the one under which the Batch 3 recipe reproduces Table 3-5 of PNNL-35884 to four decimal places on every major oxide. The code retains an explicit note against renormalizing the closure term, because folding the volatile shortfall into the oxide fractions would deflate the true waste loading.

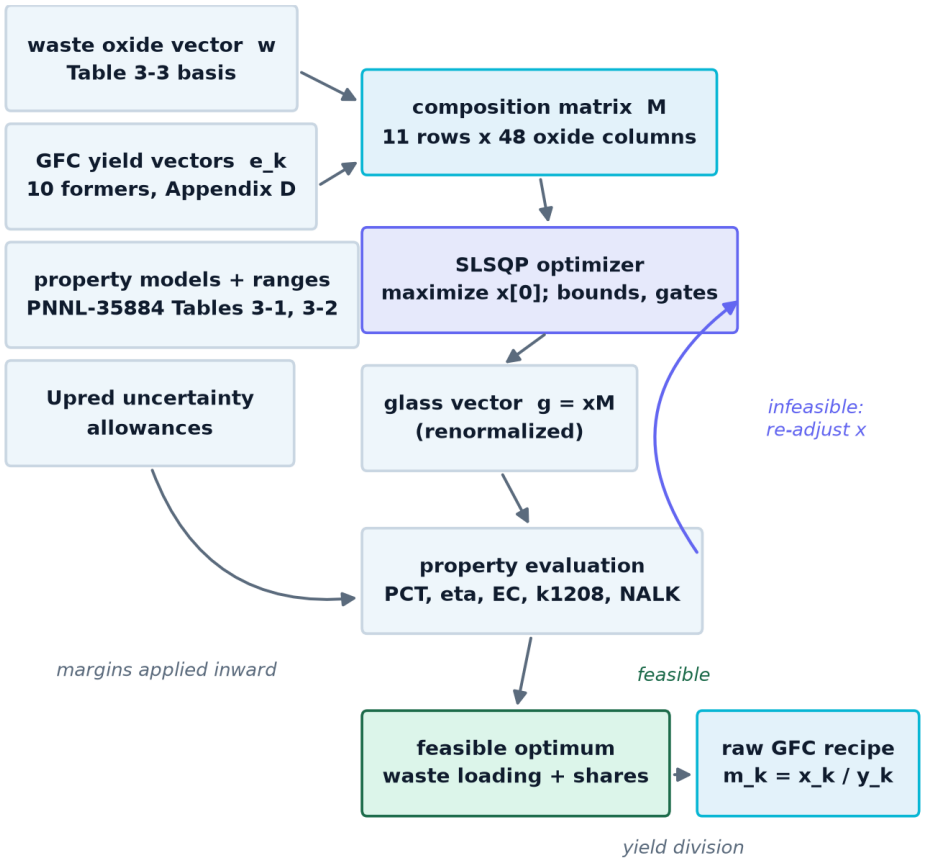


Figure 4. Data flow of the glass-former selection calculation. Solid cyan boxes denote assembly of the composition basis; the indigo box is the optimizer; green shading marks the feasible optimum and the recipe recovered from it.

Variable bounds and chemical gates

Two rules of the EWG2 logic are imposed as bounds rather than as smooth penalties. Vanadia is bounded to (0, 0) unless the waste carries more than 0.5 wt% SO3, the rationale being that the

wastes expected to benefit from sulfate-capacity enhancement are the sulfur-bearing ones. Chromia, when admissible, is bounded at 0.6 wt% of the glass. All remaining variables carry symmetric bounds (0, 1), and the component composition limits of Table 3-2 are enforced as explicit inequality constraints on the glass vector, one per bound.

Solver configuration and initialization

scipy's SLSQP is invoked with the summation equality, the fourteen inequality families, and the variable bounds, at ftol = 1e-9 and maxiter = 800. Feasibility is confirmed after each solve by evaluation of all_pass against the unpadded limits, and the best of the five starts is retained. The multi-start provision addresses a documented failure mode: wastes carrying very high soda can render the initial trial point infeasible before the line search recovers, as observed on Batch 17. Execution terminates early when the first (0.40) start yields a feasible optimum, which is the common case.

Enforcement of uncertainty

Where the property models report a predicted uncertainty Upred, the corresponding limit is enforced inward, per Equation (3). The durability limit, for instance, is imposed at 6.4368 minus its allowance, and the viscosity window at 4.0 plus and 6.0 minus their respective allowances. The distinction is material: a recipe reported feasible under outward-adjusted margins would admit a 12.87 Pa s glass as all_pass, a defect the codebase documents as removed.

$$g_j \le L_j - U_j \text{ (inward)} \quad \rightarrow \quad \eta_{1150} \le 6.0 - 0.30 = 5.70 \text{ Pa s at Batch 3}$$

Figure 5. Equation (3). Inward enforcement of a limit by its predicted-uncertainty allowance, with the Batch 3 viscosity ceiling as the worked instance.

Recovery of raw chemical demand

The solver reasons in delivered oxide; procurement requires chemical mass. After convergence, the raw demand for each selected GFC is recovered by division of its glass coordinate by its total as-fired oxide yield, per Equation (4). Coordinates below the 10^-9 threshold are treated as zero and the corresponding chemical is excluded from the recipe.

$$\hat{m}_k = \frac{x_k}{y_k}, \quad y_k = \sum_o e_{k,o} \quad \text{e.g. boric acid: } \frac{0.0856}{0.5630} = 0.1521 \text{ kg/kg glass}$$

Figure 6. Equation (4). Post-solve recovery of raw chemical mass, with the Batch 3 boric-acid entry as the worked instance.

Discussion

Determinants of former selection in general

Three waste attributes govern most of the selection behavior. First, the network former already present: a silica-rich waste admits little additional sand, whereas a silica-poor waste such as Batch 3 requires a large share. Second, the alkali already present: once the would-be glass exceeds approximately 27 wt% Na2O, the composition bound on Na2O becomes the binding constraint, beyond the influence of any GFC, and the optimizer dilutes with non-alkali formers; this behavior accounts for the 22.8% waste loading of Batch 17. Third, the sulfur gate: wastes above 0.5 wt% SO3 are offered vanadia, and the solver admitted it for all three example wastes whose sulfur lies above the gate, the model crediting V2O5 with sulfate capacity.

What the worked example establishes

Execution of the Batch 3 case by hand exhibits two features that a summary of the method conceals. The recipe is a vertex, not a compromise: four constraint margins reach zero simultaneously at the solution, which is the geometry of a maximal waste loading in a many-constraint region. Further, the identity of the selected formers turns on small margins of the waste's own composition: an additional 0.017 wt% of sulfur above the gate separates a recipe in which vanadia is admissible from one in which it is invisible. Recipes accordingly differ between tanks whose compositions differ only in trace components.

Limitations

The formulation operates on the final melter-feed composition; any pretreatment that alters the feed, such as the cyclone and overflow treatment of the companion work, must therefore run before the GFC pass, or the recipe is sized for a blend that no longer exists. The property models are valid only within their quoted composition windows, and the code treats the NSiAl and NALK windows as validity ceilings rather than as margins to be pressed against. Property coefficients are those of PNNL-35884 Rev. 1 as published; revision of that report propagates to this calculation. Finally, the solve is local: SLSQP converges to a feasible local optimum, and the multi-start mitigates, without proving the absence of, a superior vertex outside the reach of the five starts.

Conclusions

Glass-former selection is formulated as a single waste-loading maximization subject to the property constraints of PNNL-35884, and per-waste recipes are thereby produced as a byproduct of the constrained optimum rather than by lookup. The Batch 3 worked example fixes the concrete form of the result: six formers, a 54.73% waste loading, a recipe bound by the inward-adjusted durability limit with the viscosity and validity ceilings also at zero margin, and 0.5824 kg of procured chemical per kg of glass. Every numerical value in the exposition is reproducible from the code directory with the commands of Section Results.

Copyright 2026 Aspen Integra

References

[1] PNNL-35884 Rev. 1, property and formulation basis for EWG2.5 borosilicate glasses: Tables 3-1 through 3-6, and Appendix D chemical compositions.

[2] ewg_app/ewg/gfc.py, glass-forming chemical library and oxide yields of the formulation code.

[3] ewg_app/ewg/optimize.py, formulation optimizer: decision variables, constraints, multi-start SLSQP.

[4] ewg_app/ewg/properties.py, property models, model-validity ranges, and Upred allowances.

[5] ewg_app/tests.py, verification battery including the boric-acid yield round-trip and the Table 3-5 cross-check.

Copyright 2026 Aspen Integra